

<i>typevar</i> , X	type variable
<i>termvar</i> , x	term variable
<i>label</i> , l , k	field label
<i>index</i> , i , j , n , m	indices

T, S, U	$::=$		type
		X	type variable
		Top	maximum type
		$T \rightarrow T'$	type of functions
		$\forall X <: T.T'$ bind X in T'	universal type
		$\{\overline{l_i : T_i}^i\}$	record
		(T) M	
		$[X \mapsto T]T'$ M	
t	$::=$		term
		x	variable
		$\lambda x : T.t$ bind x in t	abstraction
		$t t'$	application
		$\Lambda X <: T.t$ bind X in t	type abstraction
		$t[T]$	type application
		$\{\overline{l_i = t_i}^i\}$	record
		$t.l$	projection
		let $p=t$ in t' bind $b(p)$ in t'	pattern binding
		(t) M	
		$[x \mapsto t]t'$ M	
		$[X \mapsto T]t$ M	
		σt M	
p	$::=$		pattern
		$x : T$ $b = x$	variable pattern
		$\{\overline{l_i = p_i}^i\}$ $b = b(\overline{p_i}^i)$	record pattern
v	$::=$		values
		$\lambda x : T.t$ bind x in t	abstraction
		$\Lambda X <: T.t$ bind X in t	type abstraction
		$\{\overline{l_i = v_i}^i\}$	record
Γ, Δ	$::=$		type environment
		empty	
		$\Gamma, X <: T$	
		$\Gamma, x : T$	
		$\overline{\Gamma_i}^i$ M	
σ	$::=$		multiple term substitution
		$[x \mapsto t]$	
		$\overline{\sigma_i}^i$	
<i>terminals</i>	$::=$		
		λ	
		$\rightarrow$	
		$\Rightarrow$	
		$\vdash$	
		$\longrightarrow$	
		$\forall$	

		$<:$	
		$\mapsto$	
		$\wedge$	
		$\vee$	
		$=$	
<i>formula</i>	::=	<i>judgement</i> <i>formula</i>₁ ... <i>formula</i>_n $x=x'$ $X=X'$ (<i>formula</i>) $\neg$<i>formula</i> $\forall i \in 1..m.$<i>formula</i> $\exists i \in 1..m.$<i>formula</i> <i>formula</i> $\wedge$ <i>formula</i>' $l=l'$	
<i>Judgement_in</i>	::=	$x \in \mathbf{dom}(\Gamma)$ $X \in \mathbf{dom}(\Gamma)$ $x : T \in \Gamma$ $X <: U \in \Gamma$	
<i>Jtype</i>	::=	$\Gamma \vdash \mathbf{ok}$ $\Gamma \vdash T$ $\Gamma \vdash S <: T$ $\Gamma \vdash t : T$ $\vdash p : T \Rightarrow \Delta$	type environment Γ is well-formed type T is well-formed in type environment Γ S is a subtype of T term t has type T pattern p matches type T giving bindings Δ
<i>Jop</i>	::=	$t_1 \longrightarrow t_2$ $\mathbf{match}(p, v)=\sigma$	t_1 reduces to t_2
<i>judgement</i>	::=	<i>Judgement_in</i> <i>Jtype</i> <i>Jop</i>	
<i>user_syntax</i>	::=	<i>typevar</i> <i>termvar</i> <i>label</i> <i>index</i> T t p v	

	Γ
	σ
	<i>terminals</i>
	<i>formula</i>

$$\boxed{x \in \mathbf{dom}(\Gamma)}$$

$$\frac{}{x \in \mathbf{dom}(\Gamma, x : T)} \quad \text{XING_1}$$

$$\frac{x \in \mathbf{dom}(\Gamma)}{x \in \mathbf{dom}(\Gamma, X' <: U')} \quad \text{XING_2}$$

$$\frac{x \in \mathbf{dom}(\Gamma)}{x \in \mathbf{dom}(\Gamma, x' : T')} \quad \text{XING_3}$$

$$\boxed{X \in \mathbf{dom}(\Gamma)}$$

$$\frac{}{X \in \mathbf{dom}(\Gamma, X <: U)} \quad \text{XING_1}$$

$$\frac{X \in \mathbf{dom}(\Gamma)}{X \in \mathbf{dom}(\Gamma, X' <: U')} \quad \text{XING_2}$$

$$\frac{X \in \mathbf{dom}(\Gamma)}{X \in \mathbf{dom}(\Gamma, x' : T')} \quad \text{XING_3}$$

$$\boxed{x : T \in \Gamma}$$

$$\frac{}{x : T \in \Gamma, x : T} \quad \text{TIN_1}$$

$$\frac{x : T \in \Gamma}{x : T \in \Gamma, X' <: U'} \quad \text{TIN_2}$$

$$\frac{x : T \in \Gamma}{x : T \in \Gamma, x' : T'} \quad \text{TIN_3}$$

$$\boxed{X <: U \in \Gamma}$$

$$\frac{}{X <: U \in \Gamma, X <: U} \quad \text{TIN_1}$$

$$\frac{X <: U \in \Gamma}{X <: U \in \Gamma, X' <: U'} \quad \text{TIN_2}$$

$$\frac{X <: U \in \Gamma}{X <: U \in \Gamma, x' : T'} \quad \text{TIN_3}$$

$$\boxed{\Gamma \vdash \mathbf{ok}} \quad \text{type environment } \Gamma \text{ is well-formed}$$

$$\frac{}{\mathbf{empty} \vdash \mathbf{ok}} \quad \text{GOK_1}$$

$$\frac{\Gamma \vdash T \quad \neg(x \in \mathbf{dom}(\Gamma))}{\Gamma, x : T \vdash \mathbf{ok}} \quad \text{GOK_2}$$

$$\frac{\Gamma \vdash T \quad \neg(X \in \mathbf{dom}(\Gamma))}{\Gamma, X <: T \vdash \mathbf{ok}} \quad \text{GOK_3}$$

$\boxed{\Gamma \vdash T}$ type T is well-formed in type environment Γ

$$\begin{array}{c}
\frac{\Gamma \vdash \mathbf{ok} \quad X <: U \in \Gamma}{\Gamma \vdash X} \text{GT_VAR} \\
\\
\frac{\Gamma \vdash \mathbf{ok}}{\Gamma \vdash \mathbf{Top}} \text{GT_TOP} \\
\\
\frac{\Gamma \vdash T \quad \Gamma \vdash T'}{\Gamma \vdash T \rightarrow T'} \text{GT_FUN} \\
\\
\frac{\Gamma, X <: T \vdash T'}{\Gamma \vdash \forall X <: T. T'} \text{GT_FORALL} \\
\\
\frac{\overline{\Gamma \vdash T_i^i}}{\Gamma \vdash \{ \bar{l}_i : T_i^i \}} \text{GT_RCD}
\end{array}$$

$\boxed{\Gamma \vdash S <: T}$ S is a subtype of T

$$\begin{array}{c}
\frac{\Gamma \vdash \mathbf{ok}}{\Gamma \vdash S <: \mathbf{Top}} \text{SA_TOP} \\
\\
\frac{\Gamma \vdash \mathbf{ok}}{\Gamma \vdash X <: X} \text{SA_REFL_TVAR} \\
\\
\frac{X <: U \in \Gamma \quad \Gamma \vdash U <: T}{\Gamma \vdash X <: T} \text{SA_TRANS_TVAR} \\
\\
\frac{\Gamma \vdash T_1 <: S_1 \quad \Gamma \vdash S_2 <: T_2}{\Gamma \vdash S_1 \rightarrow S_2 <: T_1 \rightarrow T_2} \text{SA_ARROW} \\
\\
\frac{\Gamma \vdash T_1 <: S_1 \quad \Gamma, X <: T_1 \vdash S_2 <: T_2}{\Gamma \vdash \forall X <: S_1. S_2 <: \forall X <: T_1. T_2} \text{SA_ALL} \\
\\
\frac{\forall i \in 1..m. \exists j \in 1..n. (k_i = l_j \wedge \Gamma \vdash S_i <: T_j)}{\Gamma \vdash \{ \bar{k}_i : S_i^{i < m} \} <: \{ \bar{l}_j : T_j^{j < n} \}} \text{SA_RCD}
\end{array}$$

$\boxed{\Gamma \vdash t : T}$ term t has type T

$$\begin{array}{c}
\frac{\Gamma \vdash \mathbf{ok} \quad x : T \in \Gamma}{\Gamma \vdash x : T} \text{TY_VAR} \\
\\
\frac{\Gamma, x : T_1 \vdash t_2 : T_2}{\Gamma \vdash \lambda x : T_1. t_2 : T_1 \rightarrow T_2} \text{TY_ABS} \\
\\
\frac{\Gamma \vdash t_1 : T_{11} \rightarrow T_{12} \quad \Gamma \vdash t_2 : T_{11}}{\Gamma \vdash t_1 t_2 : T_{12}} \text{TY_APP} \\
\\
\frac{\Gamma, X <: T_1 \vdash t_2 : T_2}{\Gamma \vdash \Lambda X <: T_1. t_2 : \forall X <: T_1. T_2} \text{TY_TABS}
\end{array}$$

$$\frac{\Gamma \vdash t_1 : \forall X <: T_{11}. T_{12} \quad \Gamma \vdash T_2 <: T_{11}}{\Gamma \vdash t_1[T_2] : [X \mapsto T_2] T_{12}} \quad \text{TY_TAPP}$$

$$\frac{\Gamma \vdash t_1 : T_1 \quad \vdash p : T_1 \Rightarrow \Delta \quad \Gamma, \Delta \vdash t_2 : T_2}{\Gamma \vdash \text{let } p=t_1 \text{ in } t_2 : T_2} \quad \text{TY_LET}$$

$$\frac{\overline{\Gamma \vdash t_i : T_i^i}}{\Gamma \vdash \{ \overline{l_i=t_i^i} \} : \{ \overline{l_i : T_i^i} \}} \quad \text{TY_RCD}$$

$$\frac{\Gamma \vdash t : \{ \overline{l_i : T_i^i} \}}{\Gamma \vdash t.l_j : T_j} \quad \text{TY_PROJ}$$

$$\frac{\Gamma \vdash t : S \quad \Gamma \vdash S <: T}{\Gamma \vdash t : T} \quad \text{TY_SUB}$$

$$\boxed{\vdash p : T \Rightarrow \Delta} \quad \text{pattern } p \text{ matches type } T \text{ giving bindings } \Delta$$

$$\overline{\vdash x : T : T \Rightarrow \mathbf{empty}, x : T} \quad \text{PAT_VAR}$$

$$\frac{\vdash p_i : T_i \Rightarrow \overline{\Delta_i^i}}{\vdash \{ \overline{l_i=p_i^i} \} : \{ \overline{l_i : T_i^i} \} \Rightarrow \overline{\Delta_i^i}} \quad \text{PAT_RCD}$$

$$\boxed{t_1 \longrightarrow t_2} \quad t_1 \text{ reduces to } t_2$$

$$\overline{(\lambda x : T_{11}. t_{12}) v_2 \longrightarrow [x \mapsto v_2] t_{12}} \quad \text{REDUCE_APPABS}$$

$$\overline{(\Lambda X <: T_{11}. t_{12})[T_2] \longrightarrow [X \mapsto T_2] t_{12}} \quad \text{REDUCE_TAPPTABS}$$

$$\frac{\mathbf{match}(p, v_1)=\sigma}{\text{let } p=v_1 \text{ in } t_2 \longrightarrow \sigma t_2} \quad \text{REDUCE_LETV}$$

$$\overline{\{ \overline{l'_i=v_i^i} \}.l'_j \longrightarrow v_j} \quad \text{REDUCE_PROJRCD}$$

$$\frac{t_1 \longrightarrow t'_1}{t_1 t \longrightarrow t'_1 t} \quad \text{REDUCE_CTX_APP_FUN}$$

$$\frac{t_1 \longrightarrow t'_1}{v t_1 \longrightarrow v t'_1} \quad \text{REDUCE_CTX_APP_ARG}$$

$$\frac{t_1 \longrightarrow t'_1}{t_1[T] \longrightarrow t'_1[T]} \quad \text{REDUCE_CTX_TYPE_FUN}$$

$$\overline{t \longrightarrow t'} \quad \text{REDUCE_CTX_RECORD}$$

$$\overline{\{ \overline{l_i=v_i^i}, l=t, \overline{l'_j=t_j^j} \} \longrightarrow \{ \overline{l_i=v_i^i}, l=t', \overline{l'_j=t_j^j} \}}$$

$$\frac{t_1 \longrightarrow t'_1}{\text{let } p=t_1 \text{ in } t_2 \longrightarrow \text{let } p=t'_1 \text{ in } t_2} \quad \text{REDUCE_CTX_LET_BINDING}$$

$$\boxed{\mathbf{match}(p, v)=\sigma}$$

$$\begin{array}{c}
\overline{\mathbf{match}(x : T, v)=[x \mapsto v]} \quad \text{M_VAR} \\
\frac{\forall i \in 1..m. \exists j \in 1..n. (l_i=k_j \wedge \mathbf{match}(p_i, v_j)=\sigma_i)}{\mathbf{match}(\{\overline{l_i=p_i}^{i < m}\}, \{\overline{k_j=v_j}^{j < n}\})=\overline{\sigma_i}^{i < m}} \quad \text{M_RCD}
\end{array}$$

Definition rules: 48 good 0 bad
 Definition rule clauses: 99 good 0 bad